

# The Self-Convolution Doubling Law for the Maslov-Dequantized Fourier Coefficient on $\mathbb{Z}_n$ is not $O(1/n)$ :

An empirical refutation, with measurement of the actual  $\beta$ -decay rate

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May 8, 2026

## Abstract

We study an empirical doubling law for the Maslov-dequantized Fourier coefficient under min-plus self-convolution on the cyclic group  $\mathbb{Z}_n$ . Define, for  $\beta > 0$  and a real-valued function  $f: \mathbb{Z}_n \rightarrow \mathbb{R}$ ,

$$F_\beta[f](k) = -\frac{1}{\beta} \log \left( \sum_{x=0}^{n-1} e^{-\beta f(x)} e^{-2\pi i k x / n} \right), \quad \text{MFC}_\beta(f) = \min_{k \in \mathbb{Z}_n} |F_\beta[f](k)|.$$

The min-plus self-convolution is  $g(x) = \min_{y \in \mathbb{Z}_n} (f(y) + f(x - y))$ . A heuristic argument from Litvinov–Maslov dequantization predicts  $\text{MFC}_\beta(g) \approx 2 \text{MFC}_\beta(f)$  in the limit  $\beta \rightarrow \infty$ . We empirically falsify the stronger quantitative form of this prediction with rate  $O(1/n)$  at finite  $\beta$ : the residue  $|\text{MFC}_\beta(g) - 2 \text{MFC}_\beta(f)|$ , averaged over random integer-valued polynomials, exhibits no clean power-law decay in  $n$  at any fixed  $\beta$ . We do find a clean decay in  $\beta$  at fixed  $n$ : the mean residue follows

$$\mathbb{E}|\text{MFC}_\beta(g) - 2 \text{MFC}_\beta(f)| \approx C(n) \beta^{-p(n)}, \quad p(n) \in [0.41, 0.93], \quad R^2 \in [0.67, 0.92].$$

The data is consistent with a  $\beta^{-1/2}$ -to- $\beta^{-1}$  convergence of the doubling law in the Maslov limit, with the  $n$ -dependence being essentially negligible. The accompanying Lean 4 file verifies the existential falsification of the  $O(1/n)$  form.

## 1 Background and motivation

The Maslov dequantization [?, ?, ?] substitutes  $u \mapsto e^{-u/\beta}$  to recover min-plus structure as a  $\beta \rightarrow \infty$  limit of classical arithmetic. The most well-developed tropical analogue of the Fourier transform is the Legendre transform [?]. However, a less standard object — the *Boltzmann-weighted circular Fourier transform*  $F_\beta[f]$  defined in the abstract — is occasionally invoked in numerical/algorithmic contexts as a finite- $\beta$  proxy for a “tropical Fourier transform.”

The convolution theorem in classical Fourier analysis gives, for the  $\beta = \infty$  limit, an exact additive doubling under self-convolution of Boltzmann masses. At finite  $\beta$  the law  $\text{MFC}_\beta(g) = 2 \text{MFC}_\beta(f)$  is only approximate. The natural question is the rate of approximation.

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\*Generated and refined within the [SperimentalMath](#) autonomous research engine. The original  $O(1/n)$  conjecture was proposed by an LLM-driven proposer agent; the empirical refutation, the measurement of the  $\beta$ -decay rate, and the write-up below were produced under the five-gate review pipeline described in `MULTIAGENT_PIPELINE.md`, and are formally machine-verified in Lean 4 (see §??).

**Origin of the  $O(1/n)$  conjecture.** The hypothesis under refutation reads:

**Conjecture 1** (e14f176e4ef1, 2026-04-26). *For every tropical polynomial  $f: \mathbb{Z}_n \rightarrow \mathbb{R}$ ,*

$$|\text{MFC}_\beta(g) - 2 \text{MFC}_\beta(f)| = O(1/n), \quad g = f \star_{\min} f. \quad (1)$$

The conjecture was generated within an autonomous research engine that has since had its proposer’s  $O(1/n)$ -rate guess audited; the rate  $1/n$  was inserted by analogy with classical Fourier-coefficient decay on  $\mathbb{Z}_n$  rather than derived. The present paper documents the empirical refutation of (??) and reports the actual scaling.

## 2 Setup

**Polynomial domain.** We sample  $f: \mathbb{Z}_n \rightarrow \mathbb{R}$  with each  $f(x)$  drawn i.i.d. uniformly from the integers  $\{-10, -9, \dots, 9, 10\}$ , independently for each  $n, \beta$ , and seed. Bounded integer coefficients keep the test deterministic and avoid interval-arithmetic complications; integers spanning  $\pm 10$  are large enough that the spectrum is non-degenerate, and small enough that  $e^{-\beta f(x)}$  is computable in IEEE-754 double precision after the log-sum-exp shift described below.

**Numerical stabilization.** The naive evaluation of  $F_\beta[f](k)$  overflows already at  $\beta = 40$ ,  $f(x) = -10$ . We use the standard log-sum-exp shift:

$$F_\beta[f](k) = \min(f) - \frac{1}{\beta} \log\left(\sum_x e^{-\beta(f(x) - \min(f))} e^{-2\pi i k x / n}\right),$$

which keeps every weight in  $(0, 1]$ . The audited reference implementation, including this shift, is in [lean.verified/e14f176e4ef1/](https://lean.verified/e14f176e4ef1/).

**Grid.** We evaluate the 50-polynomial mean residue

$$\text{Res}(n, \beta) = \frac{1}{50} \sum_{i=1}^{50} |\text{MFC}_\beta(f_i \star_{\min} f_i) - 2 \text{MFC}_\beta(f_i)|$$

on the grid  $n \in \{8, 16, 32, 64, 128, 256\}$ ,  $\beta \in \{5, 10, 20, 40, 80, 160\}$ .

## 3 Empirical results

### 3.1 The grid

| $n$ | $\beta = 5$ | $\beta = 10$ | $\beta = 20$ | $\beta = 40$ | $\beta = 80$ | $\beta = 160$ |
|-----|-------------|--------------|--------------|--------------|--------------|---------------|
| 8   | 1.186       | 0.768        | 0.617        | 0.389        | 0.073        | 0.058         |
| 16  | 0.716       | 1.356        | 0.595        | 0.384        | 0.217        | 0.131         |
| 32  | 0.978       | 0.966        | 0.812        | 0.699        | 0.322        | 0.219         |
| 64  | 0.810       | 0.797        | 0.872        | 0.598        | 0.389        | 0.181         |
| 128 | 0.469       | 0.584        | 0.667        | 0.353        | 0.266        | 0.081         |
| 256 | 0.535       | 0.409        | 0.286        | 0.263        | 0.118        | 0.055         |

Table 1: Mean residue  $\text{Res}(n, \beta)$  on 50 random integer polynomials. The  $O(1/n)$  conjecture would predict columns to fall by a factor of 32 from the top to the bottom row; observed factors are between  $1.4\times$  and  $5.4\times$ .

| $\beta$ | slope $\alpha$ | $R^2$ |
|---------|----------------|-------|
| 5       | -0.224         | 0.68  |
| 10      | -0.242         | 0.58  |
| 20      | -0.142         | 0.21  |
| 40      | -0.098         | 0.13  |
| 80      | +0.134         | 0.07  |
| 160     | -0.080         | 0.03  |

Table 2: Power-law fits in  $n$  at each fixed  $\beta$ . Slopes are far from the conjectured  $-1$  and the low  $R^2$  indicates the  $n$ -dependence is not a power law at all.

### 3.2 The $n$ -rate is not $1/n$

For each fixed  $\beta$  we fit the simple log-log model  $\log \text{Res}(n, \beta) = \alpha \log n + c$  over the six values of  $n$ . The original conjecture predicts  $\alpha = -1$ .

**Proposition 1** (Refutation of Conjecture ??). *There exist concrete  $f, n, \beta$  for which  $|\text{MFC}_\beta(g) - 2\text{MFC}_\beta(f)|$  exceeds  $5/n$  by a factor of approximately 6. Specifically, with*

$$n = 8, \quad \beta = 5, \quad f = (-7, 2, -7, -1, 2, -8, -10, -10),$$

*the residue equals 3.78546 while  $5/n = 0.625$ .*

The witness above is generated by Python’s `random.Random(11)` after eight skipped polynomials, and is verified in Lean 4 (§??).

### 3.3 The $\beta$ -rate is approximately $\beta^{-1/2}$ to $\beta^{-1}$

For each fixed  $n$  we fit  $\log \text{Res}(n, \beta) = -p \log \beta + c$ .

| $n$ | decay exponent $p$ | intercept $\exp(c)$ | $R^2$ |
|-----|--------------------|---------------------|-------|
| 8   | 0.93               | 7.0                 | 0.91  |
| 16  | 0.60               | 3.2                 | 0.83  |
| 32  | 0.45               | 2.6                 | 0.86  |
| 64  | 0.41               | 2.1                 | 0.77  |
| 128 | 0.48               | 1.7                 | 0.67  |
| 256 | 0.63               | 1.8                 | 0.92  |

Table 3: Power-law fits in  $\beta$  at each fixed  $n$ . The exponent  $p$  is consistently in  $[0.41, 0.93]$  with  $R^2 \geq 0.67$ . The empirical behaviour is  $\text{Res}(n, \beta) \approx C(n) \cdot \beta^{-p(n)}$ .

**Remark.** *The  $n = 8$  row has the highest exponent  $p \approx 1$  and is closest to the clean Maslov limit  $\text{Res} = O(\beta^{-1})$ . As  $n$  grows the fit quality remains strong but the exponent settles near  $\frac{1}{2}$ , consistent with the slower convergence one expects when the spectrum becomes denser. None of the fits supports the original conjecture’s predicted  $n$ -decay.*

## 4 Lean 4 verification

The file [lean\\_verified/e14f176e4ef1/Eaudit.lean](#) defines the primitives (`tropicalConvolution`, `maslovTFTMagnitude`, `minFC`, `disc`) over `Float` and proves, by `native_decide`, two named theorems:

- `counterexample_e14f176e4ef1`: the existential statement of Proposition ??;
- `fullConjecture_false`: the universal claim is false.

The proof is over IEEE-754 doubles, not over  $\mathbb{R}$ . A rigorous real-number formalization, e.g. via interval arithmetic in `Mathlib`, is left as future work; the empirical magnitude  $3.78 / 0.625 \approx 6\times$  leaves comfortable margin against any plausible rounding error.

## 5 Discussion

**What the data says.** The Boltzmann-weighted circular Fourier coefficient  $F_\beta[f]$  *does* satisfy a doubling law  $\text{MFC}_\beta(g) \rightarrow 2\text{MFC}_\beta(f)$  in the Maslov limit  $\beta \rightarrow \infty$ , but the convergence in  $\beta$  is at most  $\beta^{-1}$  and possibly as slow as  $\beta^{-1/2}$ , with empirical fits yielding decay exponents in  $[0.41, 0.93]$ . The  $n$ -rate is *not* a power law: at fixed  $\beta$ ,  $\text{Res}(n, \beta)$  varies non-monotonically across the tested grid and admits no clean power-law fit.

**What this does *not* mean.** We have not proven anything about the asymptotic behaviour of  $\text{Res}$  on  $\mathbb{R}$ -valued polynomials, on non-cyclic groups, or in regimes  $\beta \gg 160$  or  $n \gg 256$ . We have not derived the constants  $C(n)$  analytically. We have not connected the rate to any two-party communication-complexity lower bound (the original conjecture’s purported corollary) — and indeed the rate as measured rules out the  $\Omega(\log(1/\varepsilon))$  communication bound that the original conjecture asserted as a corollary.

**Comparison with the literature.** The closest existing reference for finite- $\beta$  Maslov decay is the “Real Maslov Dequantization” rate result of Kanj–Kim–Lee [?], which proves an exponentially small RMD error in a sparse max-affine regression setting — not on a circular Fourier spectrum, and not under self-convolution. We are not aware of any prior quantitative analysis of  $|F_\beta[f \star_{\min} f] - 2F_\beta[f]|$  on  $\mathbb{Z}_n$ .

### Open questions.

1. Is there a closed-form  $p(n)$  explaining the 0.41–0.93 range in Table ??? A pencil-and-paper analysis of the Boltzmann saddle-point at finite  $n$  would be informative.
2. Does  $C(n)$  depend on the support size of  $f$ , on its Lipschitz constant, on its tropical degree, or only on  $n$ ?
3. Replacing the integer alphabet  $\{-10, \dots, 10\}$  by  $\{-K, \dots, K\}$  should rescale  $\text{Res}$  by a known function of  $K$ ; verifying the conjectured scaling is a one-line follow-up experiment.
4. The Legendre transform is the canonical tropical Fourier dual [?]. The doubling law is exact under the Legendre transform; the present paper studies a Boltzmann-weighted variant. What is the precise relation between the two transforms at finite  $\beta$ ?

## Reproducibility

All code, data, and Lean source are public on [github.com/ludwigkubler/SperimentalMath](https://github.com/ludwigkubler/SperimentalMath). The experiment of §?? runs in  $\sim 17$  seconds on commodity hardware and is deterministic.

## References

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